Mayar Mohamed Issa

Additional Documents | International M.Sc. Data Analytics, University of Hildesheim

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Mayar Mohamed Issa
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31 October 2026

Admissions Committee
International Master in Data Analytics
University of Hildesheim

Motivation Letter: International Master in Data Analytics

Dear Members of the Admissions Committee,

I am writing to apply for the International Master in Data Analytics at the University of Hildesheim. I graduated in July 2025 with a B.Sc. in Computer Science from Pharos University in Alexandria, Excellent with Honors (CGPA 3.76 out of 4.00), and over the past year I have become sure that data analytics and machine learning are what I want to build my career on.

My interest really started in my third year, in the Machine Learning course. Until then most of my courses were about writing programs that follow rules I define myself. Seeing a model learn those rules from data instead felt completely different, and I finished the course with 94, one of my best grades. After that I paid more attention to the mathematics behind it. I did well in Numerical Computations (97), Linear Algebra (91) and Probability and Statistics (87), and courses like Data Science Methodology and Intelligent Systems showed me how much work goes into a dataset before any model is trained.

For my graduation project, Mentoro, our team of six built a mentoring platform that recommends courses, tasks and mentors to learners based on their goals and progress. I designed the full interface in Figma and built a large part of the web frontend. Working with my teammates on how the recommendations should behave, I noticed that the hard questions were never about the interface. They were about data: what we actually know about a learner, how to measure progress, and how to tell whether a recommendation is good. I could not answer those questions properly with what I knew, and that is one of the main reasons I am applying. On my own I also worked on image generation with diffusion models on CIFAR-10, which taught me how much patience training and debugging a model really takes.

After graduating, I worked for a year as a Teaching Assistant at my faculty, teaching labs in Programming 1, System Programming, Compiler Design and Human-Computer Interaction. Explaining the same idea to a full lab of students in different ways made my own understanding much more solid, and it showed me that I enjoy careful, structured work. Since August 2026 I also work as a freelance full stack developer for a small marketing agency, where part of my work is testing and comparing image and video generation models for the team. Using these models every day made me want to understand how they work from the inside, not only how to call them.

The International Master in Data Analytics fits this goal very well. I like that the program combines a strong base in machine learning and statistics with practical work on real data, and that it is closely connected to the university's machine learning research. Studying in English in an international group is also important to me, since my whole bachelor's degree was taught and assessed in English. Germany's mix of research and industry is exactly the environment where I want to grow.

After the master's, I would like to work as a data scientist or machine learning engineer, ideally on recommendation and personalisation problems, especially in education, where I have already seen how much good guidance changes a learner's path. If I find the right research topic during my studies, I would also seriously consider a PhD.

I know the program is demanding, and I am ready for that. Thank you for considering my application. I would be very happy to provide any further information.

Sincerely,

Mayar Mohamed Issa

DOCUMENT 2

AI and the Research Frontier Symposium

Faculty of Computer Science and Artificial Intelligence, Pharos University in Alexandria, 10 February 2026



DOCUMENT 3

ICPC ECPC Qualifications 2023, Honorable Mention

International Collegiate Programming Contest, 7 August 2023



Certificate of Achievement

icpc International Collegiate Programming Contest

icpc.foundation

awarded to

Mayar Mohamed

Pharos University in Alexandria

The 2023 ICPC ECPC Qualifications Collegiate Programming Contest Day 1
07 - 07 August 2023

Honorable Mention

William B. Poucher

William B. Poucher, Ph. D.
ICPC Executive Director

DOCUMENT 4

Regional Hackathon on Artificial Intelligence

Agence Universitaire de la Francophonie (AUF), Middle East, 18 March 2023



DOCUMENT 5

innovEgypt Program, Certificate of Acknowledgement

ITIDA and TIEC: completed a 45 hour training course and developed a final project for a potential startup



Networking Academy

This certificate is awarded to

Mayar Mohamed

for successfully completing

CCNA: Introduction to Networks

offered by Skills Plus Egypt
through the Cisco Networking Academy program.

Wael Samir Mohamed Abd Elghany

Wael Samir Mohamed Abd Elghany
Instructor
Skills Plus Egypt

15 Aug 2025
Completion Date

Networking Academy
CISCO

This certificate is awarded to

Mayar Mohamed

for successfully completing

CCNA: Switching, Routing, and Wireless Essentials

offered by Skills Plus Egypt
through the Cisco Networking Academy program.

Wael Samir Mohamed Abd Elghany
Instructor
Skills Plus Egypt

21 Jul 2025
Completion Date

Networking Academy

This certificate is awarded to

Mayar Mohamed

for successfully completing

CCNA: Enterprise Networking, Security, and Automation

offered by Skills Plus Egypt

through the Cisco Networking Academy program.

Wael Samir Mohamed Abd Elghany
Instructor
Skills Plus Egypt

15 Aug 2025
Completion Date

Pharos University in Alexandria

Courses Description



Program: Graduate

Faculty: Computer Science&Artificial Intelligence

Degree: BD.in Computer Science&Artificial Int.

Curriculum: Computer Science

Degree Awarded: BD.in Computer Science&Artificial Int.

Graduation Term: (2024-2025) SPRING

اسم الطالب: ميار محمد عيسي محمد سعيد

Student Name: Mayar Mohamed Issa Mohamed Said

Student ID: 202102309

National ID: 30302080202289

Nationality: EGYPT

Date of Birth: 8 Feb 2003

Academic level: Graduated

CGPA: 3.7621

Date: 8/4/2026

COURSE CODE: CS 101

COURSE_TITLE: Discrete Structures
CREDIT_HOURS: 3.000
PREREQUISITE:

COURSE_DESCRIPTION: lecture 2 & Tutorial 3
Functions- surjections- injections- inverses- composition - Relations: reflexivity- symmetry- transitivity- equivalence relations - Pigeonhole principle - Cardinality and countability - Propositional logic - Logical connectives - Truth tables - Validity - Implication- converse- inverse- negation- contradiction - Predicate logic - Universal and existential quantification - Modus ponens and modus tallness - Formal proofs - Direct proofs - Proof by counterexample - Proof by contraposition - Proof by contradiction - Well orderings - Counting arguments - Permutations and combinations - Solving recurrence relations - Trees - Undirected graphs - Directed graphs - Spanning trees

COURSE CODE: CS 103
COURSE_TITLE: Programming I
CREDIT_HOURS: 3.000
PREREQUISITE:

COURSE_DESCRIPTION: Lec 2 & Lab 2
Problem- solving strategies - Debugging strategies - Structured decomposition - concept and properties of algorithms - The role of algorithms in the problem-solving process - Basic syntax and semantics of a higher-level language - Variables, types- and assignment - Conditional and iterative control structures

COURSE CODE: DS 101
COURSE_TITLE: Fundamental of Information Systems
CREDIT_HOURS: 2.000
PREREQUISITE:

COURSE_DESCRIPTION: Lec 2 & Tut 2
Introduction to information systems by examining what is an IS, what are the different types, and how businesses use them. In addition, the module will examine the basic technologies, which provide the technical foundation for information systems. The workshops, which run parallel to the lectures, will be used to reinforce an understanding of areas covered in the lectures

COURSE CODE: MATH 101
COURSE_TITLE: Calculus I
CREDIT_HOURS: 2.000
PREREQUISITE:

COURSE_DESCRIPTION: Lec 2 & Tut 2
Functions and models, Limits and derivatives, Differentiation rules, Applications of differentiation, Integrals and applications of integration, Techniques of integration, Differential equations, Partial differential equations

COURSE CODE: PHY 101
COURSE_TITLE: Physics I



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CREDIT_HOURS: 3.000
PREREQUISITE:

COURSE_DESCRIPTION: Lec 2 & Lab 2
Newton's Laws and motion. Conservation of energy and momentum. Gravitation Electric charge in matter. Electric forces and fields. DC circuits. Waves and sound

COURSE CODE: UCS 01
COURSE_TITLE: Communication Skills (1)
CREDIT_HOURS: 1.000
PREREQUISITE:

COURSE_DESCRIPTION: Practical 2
To develop students' communication and presentation skills and train them to present themselves and their ideas orally and written in an effective manner that leads them to excellence and leadership in their work and lives.

COURSE CODE: UGB 03
COURSE_TITLE: Computer & Society
CREDIT_HOURS: 2.000
PREREQUISITE:

COURSE_DESCRIPTION: Credit Hours 2
use of personal computers and software. Effects of computer technology on contemporary society. Productivity applications, Creation of Web pages, Online behavior, Internet governance, AI, and cyber security, and societal and ethical issues in computing; privacy, security, censorship, and the changes in work, school, and entertainment fostered by computing. computer impact issues of tomorrow. the meaning of the term "computer ethics" negative impacts that computer technology might bring to individuals.

COURSE CODE: AI 102
COURSE_TITLE: Introduction to Artificial Intelligence
CREDIT_HOURS: 3.000
PREREQUISITE:

COURSE_DESCRIPTION: Lec 2 & Lab 2
Reasoning, Intelligent agents, Knowledge representation techniques, Problem solving by searching, Constraint satisfaction problems, Logic programming, Uncertain knowledge and probabilistic reasoning, Planning, Applications

COURSE CODE: CS 104
COURSE_TITLE: Programming II
CREDIT_HOURS: 3.000
PREREQUISITE: CS 103



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COURSE_DESCRIPTION: Lec 2 & lab 2
Object- class and Methods - Exception handling - Primitive types - Arrays - Data representation in memory - The concept of recursion - Inheritance and polymorphism in OOP

COURSE CODE: CS 106
COURSE_TITLE: Software Engineering
CREDIT_HOURS: 3.000
PREREQUISITE:

COURSE_DESCRIPTION: Lec 2 & lab 2
The changing pressures on software engineering practices: History of the field- definition of software- the software crisis. The Process for Developing Software and its importance The Capability Maturity Model The traditional software engineering process: The lifecycle model- evolutionary software development- incremental software development- spiral model. Prototyping Rapid software development Internet speed web-based application development -End-user development. Agile methods Extreme programming Refactoring

COURSE CODE: DS 102
COURSE_TITLE: Introduction to Data Sciences
CREDIT_HOURS: 3.000
PREREQUISITE:

COURSE_DESCRIPTION: Lec 2 & Lab 2
Data Manipulation- Data Analysis with Statistics and Machine Learning- Data Communication with Information Visualization-- Data at Scale -- Working with Big Data

COURSE CODE: MATH 102
COURSE_TITLE: Calculus II
CREDIT_HOURS: 2.000
PREREQUISITE: MATH 101

COURSE_DESCRIPTION: Lec 2 & Tut 2
Definite Integrals – Definite Integrals as Functions – The fundamental Theorem of Calculus - Integration by Parts - Integration by Rationalizing Substitutions - Powers and Products of Trigonometric Functions - Trigonometric Substitutions - Partial Fractions - Evaluating Areas in Cartesian and Polar Coordinates - Evaluating Arc Length in Cartesian and Polar Coordinates - Volumes of Revolutions by the Parallel cross section Method - Volumes of Revolutions by the Shell Method - Mathematical Induction - Roots of Polynomial Equations - Complex Numbers - Determinants and their Properties

COURSE CODE: SS 102
COURSE_TITLE: Introduction to Computer Security
CREDIT_HOURS: 3.000
PREREQUISITE:



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COURSE_DESCRIPTION: Lec 2 & Lab 2
Introduction to information security, need for security, Legal, ethical, and professional issues in information security, Risk management, Planning for security, Security technologies, and Wireless security technologies, Overview of Cryptography, Security and personnel information security maintenance

COURSE CODE: UGE 01
COURSE_TITLE: English Language (1)
CREDIT_HOURS: 2.000
PREREQUISITE: UGE 00 UGE 00 UGE 00C

COURSE_DESCRIPTION: Tutorial 2.5 & Practical 2.5
Students at this level can understand sentences and frequently used expressions related to areas of most immediate relevance (e.g. very basic personal and family information, shopping, local geography, employment). They can communicate in simple and routine tasks requiring a simple and direct exchange of information on familiar and routine matters, and can describe in simple terms aspects of his/her background, immediate environment and matters in areas of immediate need.Moreover, They can understand the main points of clear standard input on familiar matters regularly encountered in work, school, leisure, etc., and can deal with most situations likely to arise whilst travelling in an area where the language is spoken

COURSE CODE: AI 201
COURSE_TITLE: Intelligent Systems
CREDIT_HOURS: 3.000
PREREQUISITE: AI 102

COURSE_DESCRIPTION: Lec 2 & Lab 2
Definitions of intelligent systems. Optimality vs. speed tradeoff. Problem spaces. Brute-force search (DFS- BFS- uniform cost search). Heuristic search (best-first- A*- IDA*). Local search (hill-climbing- simulated annealing- genetic search). Game-playing methods (minimax search- alpha-beta pruning). Constraint satisfaction (backtracking and heuristic repair). Representation of space and time. Representations of events and actions Probabilistic reasoning. Bayes theorem. Predicate calculus and resolution. Logic programming and theorem proving.AI planning systems. Unsupervised vs. supervised learning. Inductive vs. deductive. Classification vs. clustering vs. prediction. Decision tree learning and neural network learning as examples. Design and development of knowledge-based systems. Knowledge representation mechanisms. Reasoning with uncertainty (non-monotonic logics- certainty factors- fuzzy logic). Knowledge acquisition techniques. Knowledge engineering. Tools for knowledge-based system development

COURSE CODE: CS 201
COURSE_TITLE: Data Structures and Algorithms
CREDIT_HOURS: 3.000
PREREQUISITE: CS 103

COURSE_DESCRIPTION: Lec 2 & Lab 2
Specification and representation of the elementary data types(integers- real- character- logical) - Specification- representation and implementation of basic data structure: Linear data structures (Arrays- lists- Queues- Stacks- Double ended queues- Files) - Non-linear Data Structures (Trees- Binary Trees- Graphs) - Implementation of Linked and dynamic data structures - File layout - Directories: contents and structure - Naming- searching- access- backups - Fundamental file concepts (organization- blocking- buffering) - Sequential files - Non-sequential files

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COURSE CODE: CS 202
COURSE_TITLE: Database Systems
CREDIT_HOURS: 3.000
PREREQUISITE:

COURSE_DESCRIPTION: Lec 2 & Lab 2 & tut 2
Entity-Relationship (E/R) Models - E/R Model Designs and Constraints - Object Definition Language (ODL) - The Relational Model - Converting E/R Models to Relational Designs - Converting ODL Models to Relational Designs - Design of Relational Database Schemas - 3NF and 4NF - SQL and Relational Algebra - Data Definition using SQL - Constraints in Relational Algebra and SQL - Bags vs - Sets- Extended Operators - Extended Operators and Views

COURSE CODE: CS 203
COURSE_TITLE: Logic Circuit Design
CREDIT_HOURS: 3.000
PREREQUISITE: PHY 101

COURSE_DESCRIPTION: Lec 2 & Lab 2
Provides a modern introduction to logic design and the basic building blocks used in digital systems, in particular digital computers. It starts with a discussion of combinational logic: logic gates, minimization techniques, arithmetic circuits, and modern logic devices such as field programmable logic gates. The second part of the course deals with sequential circuits: flip-flops, synthesis of sequential circuits, and case studies, including counters, registers, and random-access memories. State machines in complex systems using programmable logic devices. Different representations including truth table, logic gate, timing diagram, switch representation, and state diagram.

COURSE CODE: MATH 201
COURSE_TITLE: Probability and Statistics
CREDIT_HOURS: 2.000
PREREQUISITE: MATH 101

COURSE_DESCRIPTION: Lec 2 & Tut 2
Elementary introduction to probability and statistics with applications. Topics include basic combinatory- random variables- probability distributions- Bayesian inference- hypothesis testing- confidence intervals- linear regression

COURSE CODE: UEB 01
COURSE_TITLE: Entrepreneurship & Innovation
CREDIT_HOURS: 2.000
PREREQUISITE:

COURSE_DESCRIPTION: Credit Hours 2
Definitions and evolution of entrepreneurship and innovation. Role of innovation in entrepreneurship and organizational development. Innovation as a capability/ core competence. Understanding entrepreneurial and innovative individuals. Role of innovation in new growth firm creation (independent entrepreneurship). Role of innovation in established enterprise development (corporate entrepreneurship). Role of innovation in high growth or gazelle firms. Role of technology in innovation. Role of champions and culture in innovation. Implementing innovation. Measuring innovation performance. Some key issues in innovation management.

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COURSE CODE: UGE 02
COURSE_TITLE: English Language (2)
CREDIT_HOURS: 2.000
PREREQUISITE: UGE 01 UGE 01

COURSE_DESCRIPTION: Tutorial 2.5 & Practical 2.5
Students at this level can understand the main points of clear standard input on familiar matters regularly encountered in work school, leisure, etc. They can deal with most situations likely to arise whilst travelling in an area where the language is spoken, produce simple connected text on topics which are familiar or of personal interest. And they can describe experiences and events, dreams, hopes & ambitions and briefly give reasons and explanations for opinions and plans.

COURSE CODE: CS 204
COURSE_TITLE: Computer Architecture
CREDIT_HOURS: 3.000
PREREQUISITE:

COURSE_DESCRIPTION: Lec 2 & Lab 2
Fundamentals of processor, memory, and network design, advanced processor and memory design. Example architecture from industry.

COURSE CODE: CS 205
COURSE_TITLE: Algorithms Design
CREDIT_HOURS: 3.000
PREREQUISITE: CS 201

COURSE_DESCRIPTION: Lec 2 & Lab 2
Asymptotic notations, solving recursive relations, Basic analysis measures: Worst and average-case complexity bounds, Amortized analysis, Randomization, Fundamental design strategies: Divide-and-conquer, Dynamic programming, and Greedy methods, String algorithms, Geometric algorithms, Number-theoretic algorithms, Complexity classes, NP complete problems, Approximation algorithms

COURSE CODE: CS 206
COURSE_TITLE: Operating System
CREDIT_HOURS: 3.000
PREREQUISITE:

COURSE_DESCRIPTION: Lec 2 & Lab 2
Structuring methods and the layered model - Evolution of hardware/software techniques - Device organization - Interrupts - User/system state and protection - State diagrams - Structures (ready list- process control blocks- and so forth) - Dispatching and context switching - Concurrent execution - Mutual exclusion Deadlock and starvation - Models and mechanisms (semaphores- monitors- rendezvous) - Producer-consumer problems - Preemptive and non-preemptive scheduling - Schedulers and policies - Processes and threads - Deadlines and real-time issues - Overlays- swapping- and partitions - Paging and segmentation - Memory mapped files - Placement and replacement policies - Serial or parallel device - Servers and interrupts - File layout - Directories: contents and structure - Sequential files - Non sequential files - Process and task scheduling - Memory and disk management - Real-time systems



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COURSE CODE: CS 401
COURSE_TITLE: Mobile Application
CREDIT_HOURS: 3.000
PREREQUISITE:

COURSE_DESCRIPTION: Lec 2 & Lab 2
Android platform and the Android Studio IDE. Android components- Activities and their lifecycle- Intents UI Design: Widgets and Layouts- UI Events- Event Listeners 2D graphics: Canvas/Drawing using a view Multimedia: Audio playback and Media Player Network support

COURSE CODE: MATH 202
COURSE_TITLE: Linear Algebra
CREDIT_HOURS: 2.000
PREREQUISITE: MATH 102

COURSE_DESCRIPTION: Lec 2 & Tut 2
Matrix theory and linear algebra. Systems of equations- vector spaces- determinants- eigenvalues- similarity- positive definite matrices. Linear operators on finite dimensional vector spaces. The algebraic co-ordinate free methods throughout the course. The scalar field will be arbitrary (even a finite field)- however- in the treatment of vector spaces with inner products- special attention will be given to the real and complex cases. Determinants via the theory of multilinear forms. Variety of examples of the important concepts

COURSE CODE: UEB 02
COURSE_TITLE: Critical Thinking
CREDIT_HOURS: 2.000
PREREQUISITE:

COURSE_DESCRIPTION: Credit Hours 2
Fundamental concepts of critical thinking (logic, argument analysis, rhetoric, reasoning with probabilities, the importance of background knowledge, etc.). The role that critical thinking principles play in good essay writing. how our tribal psychology and political polarization affect our ability to think critically. The importance of critical thinking for personal development, participation in democratic society, and the pursuit of wisdom. how cognitive biases make us prone to errors in how we form beliefs and make judgments. how debates in science often turn on the meanings of key terms, like "theory", "fact", "hypothesis" and "law".

COURSE CODE: UGE 03
COURSE_TITLE: English Language (3)
CREDIT_HOURS: 2.000
PREREQUISITE: UGE 02 UGE 02



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COURSE_DESCRIPTION: Tutorial 2.5 & Practical 2.5
Students at this level can understand the main points of clear standard input on familiar matters regularly encountered in work, school, leisure, etc. They can deal with most situations likely to arise whilst travelling in an area where the language is spoken. Also, they can produce simple connected text on topics which are familiar or of personal interest, and can describe experiences and events, dreams, hopes & ambitions and briefly give reasons and explanations for opinions and plans.

COURSE CODE: AI 301
COURSE_TITLE: Machine Learning
CREDIT_HOURS: 3.000
PREREQUISITE: AI 201

COURSE_DESCRIPTION: Lec 2 & Lab 2
Feature representation, Similarity, Dimensionality reduction, Supervised learning, Regression, Unsupervised learning, Evaluation, Fundamentals of neural networks (Feedforward and Backpropagation), Reinforcement learning, Applications

COURSE CODE: CS 301
COURSE_TITLE: Computer Networks
CREDIT_HOURS: 3.000
PREREQUISITE:

COURSE_DESCRIPTION: Lec 2 & Lab 2
Introduction to network architecture, layering, and protocols, Principles of network applications and application layer protocols examples, Socket programming, Introduction to transport layer protocols: Principles of reliable data delivery (error control, congestion control, and flow control), TCP protocol, Introduction to the network layer: Network layer addressing, Routing and forwarding, Principles of routing algorithms

COURSE CODE: CS 305
COURSE_TITLE: Web programming
CREDIT_HOURS: 3.000
PREREQUISITE: CS 103

COURSE_DESCRIPTION: Lec 2 & Lab 2
Web fundamentals, Programming Languages for the Web, HTML Basics, Using HTML, Using CSS and templates, Basics of JavaScript, Programming with JavaScript, Introduction to front end programming, PHP, Django, AngularJS, React, responsive web design, Full stack state management, Security pitfalls and basic solutions. Cloud computing. Cloud Service Providers. Cloud Computing Architecture. Cloud Computing Terminology. Communication-as-a-Service (CaaS). Infrastructure-as-a-Service (IaaS). Monitoring-as-a-Service (MaaS). Software-as-a-Service (SaaS). Platform-as-a-Service (PaaS).

COURSE CODE: CS 307
COURSE_TITLE: Theory of computation
CREDIT_HOURS: 3.000
PREREQUISITE: CS 101



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COURSE_DESCRIPTION: Lec 2 & Tut 3
Deterministic Finite Automata (DFA) and Regular Languages - Non-deterministic Finite Automata (NFA) - Equivalence of NFAs and DFAs - Closure Properties - Regular Expressions - The Pumping Lemma - Pushdown Automata - Pushdown Automata - Context Free Languages - Ambiguity and Chomsky Normal Form - Turing Machines - Turing Machines Variants - Enumerators - Decidability - Diagonalization - The Halting Problem - Reducibility - Time Complexity - P and NP - Poly-time reductions - NP-Completeness

COURSE CODE: MATH 301
COURSE_TITLE: Numerical Computations
CREDIT_HOURS: 2.000
PREREQUISITE: MATH 202

COURSE_DESCRIPTION: Lec 2 & Tut 2
Review of Mat lab Programming Language - Additional DATA Types – Derived DATA Types - Debugging Mat lab Programs – Procedures Applications: Interpolation - Numerical Integration and differentiation - Solution of Non-linear Equations - Solution of Systems of Linear Equations - Initial Value Problems - Boundary Value Problems - Elements of Grid Methods for Partial Differential Equations

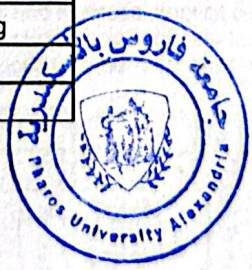
COURSE CODE: SS 402
COURSE_TITLE: Block chain computing
CREDIT_HOURS: 3.000
PREREQUISITE:

COURSE_DESCRIPTION: Lec 2 & Lab 2
Introduction to block chains and cryptocurrencies, Distributed ledger technologies, Trust and vulnerabilities, Consensus mechanisms, Basics of contract law, Smart contracts, Regulations and laws of block chains, Security of block chains

COURSE CODE: AI 302
COURSE_TITLE: Robotics
CREDIT_HOURS: 3.000
PREREQUISITE: AI 301

COURSE_DESCRIPTION: Lec 2 & Lab 2
Foundations and principles of robotic kinematics, Transformations, Forward kinematics, Inverse kinematics, Differential kinematics (Jacobians), Manipulability and basic equations of motion, Programming on robot arms, Applications, localization and mapping, Kalman filters

COURSE CODE: CS 207
COURSE_TITLE: System simulation and modelling
CREDIT_HOURS: 3.000
PREREQUISITE: CS 101



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COURSE_DESCRIPTION: Lec 2 & Lab 2
Systems Modeling Concepts- Modeling formalisms and their simulators- Cellular automata- Discrete event systems modeling- Continuous system modeling- Bond Graph approach to modeling physical systems- Stochastic systems- Hybrid (continuous + discrete) systems - Abstraction models- Distributed systems- verification and validation mechanisms

COURSE CODE: CS 209
COURSE_TITLE: Advanced Programming
CREDIT_HOURS: 3.000
PREREQUISITE: CS 104

COURSE_DESCRIPTION: Lec 2 & Lab 2
Object-oriented application programming. Applications such as standalone, web, enterprise solutions. Most in-demand programming languages, a standard for enterprise software, web-based content, games and mobile apps, Android operating system. Multiple software platforms, program written on Mac OS X, for example, could also run in a Windows environment.

COURSE CODE: CS 304
COURSE_TITLE: System Programming
CREDIT_HOURS: 3.000
PREREQUISITE: CS 206

COURSE_DESCRIPTION: Lec 2 & Lab 2
System calls and how they interact with the kernel. File manipulation looking at how to create, delete, and edit files using system call. Memory management and how exactly Unix deals with virtual memory addressing. Processes: Threads. Synchronization issues that arise when using processes and threading

COURSE CODE: CS 308
COURSE_TITLE: Computer Graphics
CREDIT_HOURS: 3.000
PREREQUISITE: MATH 301

COURSE_DESCRIPTION: Lec 2 & Lab 2
Raster and vector graphics system - Video display devices - Physical and logical input devices - Issues facing the developer of graphical systems - Hierarchy of graphics software - User interface - Half tones - Font generation: outline vs - bitmap - Representation of polyhedral objects - Scan conversion of 2D primitive- forward differencing - Tessellation of curved surfaces - Homogeneous coordinates - Affine transformations (scaling- rotation- translation) - Viewing transformation. Clipping. Hidden surface removal methods. Z-buffer and frame buffer- color channels (a channel for opacity). Color models (RGB- HIVS- CYM). Light source properties - material properties - ambient- diffuse- specular reflections. Reflection model. Rendering of a polygonal surface- flat shading- Texture mapping- bump texture- environment map. Ray tracing. Image synthesis- sampling techniques- Parametric polynomial curves and surfaces. Implicit curves and surfaces. Bezier curves and surfaces- control points- B-spline curves and surfaces- local editing- control points. NURBS curves and surfaces. Constructive Solid Geometry for solid modeling. Boundary Representation of solids (B-Rep).

COURSE CODE: SS 302
COURSE_TITLE: Internet of Things



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CREDIT_HOURS: 3.000
PREREQUISITE: CS 301

COURSE_DESCRIPTION: Lec 2 & Lab 2
Fundamentals of the Internet of Things (IoT), Impact of IoT on society, Business process in IoT, the design and implementation of IoT devices as embedded systems, Hardware and software components for IoT, IoT node authentication, Developing smart cities, IoT security

COURSE CODE: CS 310
COURSE_TITLE: Human Computing Interaction
CREDIT_HOURS: 3.000
PREREQUISITE:

COURSE_DESCRIPTION: Lec 2 & Lab 2
Mapping - Affordances - Constraints - Seven Stages of Action - Schneiderman's 8 Golden Rules - Information Visualization - Model Human Processor - Keystroke Level Model - Fit's law - Input devices (Keyboard- Pointing- Voice) - Output devices (Displays- Color- Sound) - Interaction Styles (direct manipulation- menu selection- form-fill-in- command languages) - Windows - Icons - Menus - Dialogue Boxes - Concepts (grids- simplicity- consistency- white space) - Context Sensitive Help - Tutorials - Reference Material - Cognitive Walkthrough - Heuristic Evaluation - Expert Reviews - Controlled Experiments (subjects- dependent & independent variables- statistics) - Synchronous / Asynchronous tools - Audio / Video - Shared Workspaces

COURSE CODE: CS 403
COURSE_TITLE: Advanced computer networks
CREDIT_HOURS: 3.000
PREREQUISITE: CS 301

COURSE_DESCRIPTION: Lec 2 & Lab 2
Principles, architectures, and protocols used in modern networked systems, such as the Internet, wireless and mobile networks, high performance networks and data center networks. Existing technology in building large, complex networked systems, and provide concrete experience of the challenges through a series of lab exercises. Wireless networks and mobility issues at the network and transport layer (Mobile IP and micro mobility protocols, TCP in wireless environments). Data center and high-performance networking. Network virtualization.

COURSE CODE: CS 405
COURSE_TITLE: Distributed Systems and Cloud Computing
CREDIT_HOURS: 3.000
PREREQUISITE: CS 301

COURSE_DESCRIPTION: Lec 2 & Lab 2
Introduction to Cloud Computing, Cloud computing platforms and infrastructure, Parallel programming in the Cloud, Distributed storage systems, Virtualization, Cloud security, Cloud performance

COURSE CODE: CS 491
COURSE_TITLE: Project I



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CREDIT_HOURS: 3.000
PREREQUISITE: CS 304

COURSE_DESCRIPTION: Lec 2 & Lab 2
Demonstrate the skills they have learned in an applications area of their own choosing. projects must include a research component

COURSE CODE: DS 411
COURSE_TITLE: Healthcare Database systems
CREDIT_HOURS: 3.000
PREREQUISITE:

COURSE_DESCRIPTION: Lec 2 & Lab 2
The course aims to introduce the principles of health care organization and policy in a comparative perspective nationally and internationally, identify the key characteristics and components of health care systems, assess each health care system's strengths and weaknesses, and discuss the recent health care reform efforts

COURSE CODE: UCS 02
COURSE_TITLE: Communication Skills (2)
CREDIT_HOURS: 1.000
PREREQUISITE: UCS 01

COURSE_DESCRIPTION: Practical 2
To provide a professionally-oriented, skills-based learning that helps the students to get appropriate jobs and emphasizes on providing the community with qualified graduates show excellence in their career.

COURSE CODE: CS 309
COURSE_TITLE: Embedded Systems
CREDIT_HOURS: 3.000
PREREQUISITE: CS 201

COURSE_DESCRIPTION: Lec 2 & Lab 2
Introduction to computer architecture, Sensors, Actuators, Input-Output (I/O) modeling, I/O Interfacing, Microcontrollers, Real-time operating systems, Energy-aware design, Case studies and applications of embedded control systems, Recent trends and developments

COURSE CODE: CS 311
COURSE_TITLE: Software Quality Assurance and Testing
CREDIT_HOURS: 3.000
PREREQUISITE: CS 106



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COURSE_DESCRIPTION: Lec 2 & Lab
Success, Reflections, Terms and Concepts. Competencies, Standards, Software Lifecycle Models, Software Quality Assurance. Software Project Management, Adoption of PM Practices. Reviews and Inspections, Software Testing, Testing Effort, Advances in Testing. Software Quality as a Profession. Metrics and Analysis, Software Configuration Management Release Management.

COURSE CODE: CS 404
COURSE_TITLE: Compiler and Language Theory
CREDIT_HOURS: 3.000
PREREQUISITE: CS 304

COURSE_DESCRIPTION: Lec 2 & Lab 2
Principles and practices for design and implementation of compilers. Specific topics include lexical analysis; parsing theory; symbol tables; type systems; scope; semantic analysis; intermediate representations; runtime environments; code generation; and basic program analysis and optimization. Students construct a compiler for a simple object-oriented language during course programming projects

COURSE CODE: CS 492
COURSE_TITLE: Project II
CREDIT_HOURS: 3.000
PREREQUISITE: CS 491

COURSE_DESCRIPTION: Lec 2 & Lab 2
Demonstrate the advanced skills they have learned in an applications area of their own choosing in project1. projects must include a research component

COURSE CODE: DS 302
COURSE_TITLE: Data Science Methodology
CREDIT_HOURS: 3.000
PREREQUISITE: DS 102

COURSE_DESCRIPTION: Lec 2 & Lab 2
The data science framework, Business understanding, RFM modelling, Data collection and preparation, Structured and unstructured data, Cleaning and Exploration, Data visualization, Market basket analysis, Modeling, Evaluation, Feedback, Use cases

COURSE CODE: DS 307
COURSE_TITLE: Bioinformatics Programming and Scripting
CREDIT_HOURS: 3.000
PREREQUISITE: CS 201



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COURSE_DESCRIPTION: Lec 2 & Lab 2
Algorithms for solving various biological problems with a handful of programming challenges helping implement these algorithms in Python. offers a gently paced Introduction to Bioinformatics Specialization





